

**PREDICTING STUDENT EXAM PERFORMANCE:
A REGRESSION-BASED ANALYSIS OF BEHAVIOURAL, FAMILIAL, AND
SCHOOL-RELATED FACTORS AFFECTING ACADEMIC OUTCOMES**

**Project Report Submitted in fulfillment of the requirements for the
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ABSTRACT

Student academic performance is shaped by a wide range of behavioural, familial, and institutional factors, yet many of these influences are rarely examined together in a single quantitative framework. This project analyses the Student Performance Factors dataset, comprising 6,607 student records and 20 variables covering study habits, attendance, parental involvement, access to resources, and related factors, with the goal of understanding and predicting student Exam Score. The dataset was cleaned by removing duplicate records, imputing missing categorical values with the column mode, and correcting an implausible outlier value in the target variable. Exploratory and statistical analysis showed that Attendance ($r = 0.581$) and Hours Studied ($r = 0.446$) were the variables most strongly associated with Exam Score, while sleep hours and physical activity showed negligible linear association. Categorical variables such as Parental Involvement and Access to Resources were ordinally encoded, nominal variables were one-hot encoded, and an Engagement Index and Study-to-Sleep Ratio were engineered as additional predictors. A Linear Regression model and a Random Forest Regressor were trained on an 80/20 train-test split. The Linear Regression model achieved the stronger generalisation performance, with a test R-squared of 0.7701 and a Root Mean Squared Error of 1.803 points, compared with an R-squared of 0.65 for the Random Forest model. These results indicate that a large share of the variation in exam performance can be explained by a compact set of behavioural and contextual factors, with attendance and study engagement emerging as the strongest levers. The findings suggest practical, low-cost interventions - such as attendance monitoring and structured study-support programs - that schools could adopt to improve student outcomes.

KEYWORDS

ExploratoryData Analysis; Data Preprocessing; Regression Analysis; Feature Engineering; Predictive Analytics; Student Performance

TABLE OF CONTENTS

1. Introduction.....	5
2. Objectives and Research Questions.....	5
3. Dataset Description.....	6
4. Data Cleaning.....	7
5. Data Preprocessing.....	8
6. Exploratory Data Analysis.....	9
7. Data Visualisation.....	10
8. Statistical Analysis.....	16
9. Machine Learning Methodology.....	17
10. Model Results and Evaluation.....	18
11. Discussion.....	21
12. Limitations.....	21
13. Conclusion and Recommendations.....	21
14. References.....	22
15. Appendix.....	22

1. INTRODUCTION

1.1 Background

Academic performance is one of the most closely watched outcomes in education systems worldwide, influencing student opportunities, institutional rankings, and public policy. While traditional educational research has long recognised that performance is multi-causal, most classroom-level decisions are still made on intuition rather than data. The growing availability of structured student-level datasets, capturing study habits, attendance, family background, and school characteristics, creates an opportunity to apply data science techniques to identify which factors most strongly influence outcomes, and to what extent those outcomes can be predicted in advance.

1.2 Context of the Problem

Teachers, school administrators, and parents are the primary stakeholders who could benefit from a clearer, quantitative understanding of what drives exam performance. In practice, resources for interventions such as tutoring, counselling, or attendance monitoring are limited, so knowing which factors carry the most predictive weight helps target these resources more effectively. A predictive model built on historical student data can also serve as an early-warning tool, flagging students whose combination of study habits and circumstances puts them at risk of underperforming.

1.3 Problem Statement and Motivation

Given a set of behavioural, familial, and institutional factors describing a student, can we accurately estimate that student's final exam score, and which of these factors contribute most to the outcome? Educators often lack a systematic, data-driven way to identify which students are likely to underperform and why. A regression-based approach that both predicts the outcome and quantifies the relative importance of each input factor can provide actionable, evidence-based guidance for designing targeted academic support programs, rather than relying on generic, one-size-fits-all interventions.

1.4 Scope of the Study

This study uses a single cross-sectional dataset of student records containing study, attendance, and demographic/contextual variables, along with a final Exam Score. The project covers data cleaning, exploratory data analysis, statistical analysis, feature engineering, and the development and evaluation of regression models to predict Exam Score. It does not cover longitudinal tracking of the same students over time, does not incorporate psychometric or classroom-observation data, and does not attempt to establish causal relationships between the study variables and performance; all relationships reported are associative, not causal. The primary objectives and prediction question that follow (Section 2) define the boundaries of the analysis in more detail.

2. OBJECTIVES AND RESEARCH QUESTIONS

2.1 Objectives

- To understand and prepare the Student Performance Factors dataset for analysis, including cleaning missing values and identifying data quality issues.
- To identify important patterns, relationships, and factors associated with student Exam Score through exploratory and statistical analysis.

- To engineer meaningful derived features (such as an engagement index) that may improve predictive power.
- To build and compare regression models (Linear Regression and Random Forest) to predict Exam Score.
- To evaluate model performance using appropriate regression metrics and identify the most influential predictors.
- To translate the results into practical, evidence-based recommendations for improving student outcomes.

2.2 Research Questions

1. What are the major characteristics and data quality issues in the Student Performance Factors dataset?
2. Which variables are most strongly associated with Exam Score?
3. How does Exam Score vary across categorical groups such as parental involvement and peer influence?
4. How accurately can a regression model predict Exam Score using the available features, and which model generalises better? (Primary prediction question)

3. DATASET DESCRIPTION

3.1 DataSource

The dataset used in this project, 'Student Performance Factors' (P_2_StudentPerformanceFactors.csv), was provided as one of the ten approved capstone project datasets in the LMS Study Material section for the IDRA Data Science & AI Training Program. It contains anonymised, student-level records with no personally identifiable information, representing a single cross-sectional snapshot (no explicit time period is attached to individual records).

3.2 Dataset Size and Structure

The raw dataset contains 6,607 rows (student records) and 20 columns. Nineteen columns are candidate predictors, a mix of numeric behavioural measures (e.g. Hours Studied, Attendance, Sleep Hours) and categorical contextual variables (e.g. Parental Involvement, School Type, Peer Influence).

Attribute	Value	Attribute	Value
Total records	6607	Total columns	20
Numeric columns	7	Categorical columns	13
Duplicate rows found	0	Columns with missing values	3

Table 1. Summary of the Dataset

3.3 DataDictionary

The table below summarises the main variables used in the analysis, their type, and role.

Variable	Type	Description	Role
Hours_Studied	Numeric	Weekly hours spent studying	Feature
Attendance	Numeric (%)	Class attendance percentage	Feature

Variable	Type	Description	Role
Sleep_Hours	Numeric	Average hours of sleep per night	Feature
Previous_Scores	Numeric	Score on a prior assessment	Feature
Tutoring_Sessions		Number of tutoring sessions attended	Feature
Parental_Involvement	Numeric	Level of parental involvement (Low/Medium/High)	Feature
Access_to_Resources	Categorical	Access to learning resources (Low/Medium/High)	Feature
Peer_Influence	Categorical	Nature of peer influence (Negative/Neutral/Positive)	Feature
School_Type	Categorical	Public or Private school	Target
Exam_Score	Categorical	Final exam score achieved by the student	Target
	Numeric		

Table 2. Data Dictionary (Selected Variables)

3.4 Target Variable

Exam_Score is the target variable (y) for this project: a continuous numeric score, which makes this a regression problem. All other 19 columns are conceptually treated as candidate predictors (X) and are considered independent of the target at collection time — none of them are derived from or leak information about the final exam score, so no target-leakage variables were excluded.

4. DATA CLEANING

Before analysis, the raw dataset was examined for missing values, duplicate records, inconsistent categories, incorrect data types, and outliers.

4.1 Missing Values

Three categorical columns contained missing values: Teacher_Quality, Parental_Education_Level, and Distance_from_Home. In each case the proportion of missing data was small (under 1.4% of records), so listwise deletion was avoided to prevent unnecessary loss of data and possible bias from dropping rows. Missing values were instead imputed with the column mode (the most frequent category), which is an

appropriate and low-bias treatment for low-missingness categorical variables.

Column	Missing Count	% Missing	Treatment
Teacher_Quality	78	1.18%	Imputed with column mode (most frequent category)
Parental_Education_Level	90	1.36%	Imputed with column mode (most frequent category)
Distance_from_Home	67	1.01%	Imputed with column mode (most frequent category)

Table 3. Missing Value Summary and Treatment

4.2 Duplicate Records

A check for fully duplicated rows found 0 duplicate records in the dataset, so no rows were removed on this basis. The dataset size therefore remained unchanged after this step.

4.3 Data Types and Inconsistent Values

All numeric columns were already stored with appropriate numeric data types, and all categorical columns used a small, consistent set of text labels (e.g. 'Low' / 'Medium' / 'High') with no spelling variants or case-inconsistencies detected. No data type conversions were required.

4.4 Outlier Handling

Outliers in the numeric columns were checked using the interquartile range (IQR) method (values beyond $1.5 \times \text{IQR}$ from the first/third quartiles were flagged). Tutoring_Sessions and Hours_Studied showed a number of statistically flagged high values, but these are plausible real-world behaviours (some students attend many tutoring sessions or study long hours) and were retained rather than removed. The target variable, Exam_Score, contained 1 record(s) with a value above the maximum possible score of 100, which is not a plausible outcome and was therefore treated as a data entry artefact; this value was capped (winsorised) at 100 rather than deleted, preserving the record while correcting the impossible value.

Numeric Variable	IQR Outliers (count)	Lower Bound	Upper Bound
Hours_Studied	43	4	36
Attendance	0	40	120
Sleep_Hours	0	3	11
Previous_Scores	0	25.5	125.5
Tutoring_Sessions	430	-0.5	3.5
Physical_Activity	0	-1	7
Exam_Score	104	59	75

Table 4. Outlier Detection Summary (IQR Method)

4.5 Validation: Before vs After Cleaning

To confirm the cleaning steps resolved the identified issues, the table below compares key data-quality indicators before and after cleaning.

Metric	Before Cleaning	After Cleaning	Change
Rows	6607	6607	0
Columns	20	20	0
Total missing values	235	0	Resolved
Max Exam Score	101	100	Capped at 100

Table 5. Data Quality Before and After Cleaning

5. DATA PREPROCESSING

5.1 Categorical Encoding

Categorical variables with a natural order (Parental_Involvement, Access_to_Resources, Motivation_Level, Family_Income, Teacher_Quality — each Low/Medium/High) were ordinal-encoded to values 1, 2, and 3, preserving their inherent ranking. Nominal categorical variables with no natural order (e.g. School_Type, Peer_Influence, Gender, Internet_Access, Learning_Disabilities, Distance_from_Home, Parental_Education_Level, Extracurricular_Activities) were one-hot encoded using pandas get_dummies with drop_first=True to avoid multicollinearity (the dummy variable trap).

5.2 Feature Scaling and Transformation

Numeric features were standardised (zero mean, unit variance) using scikit-learn's StandardScaler before being passed to the Linear Regression model, since linear models with comparably-scaled coefficients are easier to interpret. The Random Forest model, being tree-based and scale-invariant, was trained on the unscaled feature set.

5.3 Feature Engineering

Three derived features were created to capture combined behavioural effects not represented by any single original column:

Feature	Formula / Definition	Rationale	Type
Engagement_Index	Hours_Studied + Attendance/10 + Tutoring_Sessions x 2	Combines the three strongest engagement-related predictors into a single composite score	Numeric
Study_Sleep_Ratio	Hours_Studied / Sleep_Hours	Captures whether study time comes at the expense of adequate sleep	Numeric
High_Attendance	1 if Attendance >= median, else 0	Simple binary flag distinguishing above- and below-median attenders	Binary

Table 6. Engineered Features

5.4 Train-Test Split

The cleaned and encoded dataset was split into training and test sets using an 80/20 ratio (test_size = 0.20) with a fixed random_state = 42 for reproducibility. This resulted in 5285 training records and 1322 test records. The scaler was fit only on the training data and then applied to the test data, to avoid data leakage from the test set into the preprocessing step.

5.5 Final Feature Set

After encoding and feature engineering, the final feature set contained 25 input variables (a mix of original numeric features, ordinal-encoded categorical features, one-hot encoded nominal features, and engineered features), with Exam_Score as the continuous target variable.

6. EXPLORATORY DATA ANALYSIS

6.1 Dataset Overview

The cleaned dataset consists of 6,607 records across 20 original columns (before encoding), split between 7 numeric and 13 categorical variables. Basic descriptive statistics for the numeric variables are presented in Section 8 (Statistical Analysis); this section focuses on the relationships between individual variables and the target, which directly inform the feature choices used in modelling.

6.2 Relationships with the Target Variable

The correlation of each numeric predictor with Exam Score was computed to identify the variables most worth exploring further and most likely to matter for prediction.

Predictor	Correlation with Exam Score
Attendance	0.581
Hours_Studied	0.446
Previous_Scores	0.175
Tutoring_Sessions	0.156
Physical_Activity	0.028
Sleep_Hours	-0.017

Table 7. Correlation of Numeric Predictors with Exam Score

Attendance and Hours Studied are clearly the two numeric variables most strongly associated with Exam Score, followed by Previous Scores and Tutoring Sessions. Sleep Hours and Physical Activity show correlations very close to zero, suggesting limited direct linear association with performance within the range observed in this dataset. For the key categorical variables, mean Exam Score was compared across group levels: mean scores rise consistently from Low to Medium to High Parental Involvement and from Negative to Neutral to Positive Peer Influence (see the boxplots in Section 7, Figures 2 and 3), indicating that both family involvement and social environment are associated with better outcomes, alongside the individual behavioural factors.

6.3 What the EDA Tells Us About Modelling Choices

These exploratory findings directly shaped the modelling approach: Attendance, Hours Studied, Previous Scores and Tutoring Sessions were retained as core numeric predictors; the ordered categorical variables (Parental Involvement, Access to Resources, etc.) were preserved as ordinal features rather than being dropped, since their group-level averages show a clear monotonic trend; and the correlation matrix (Figure 6, Section 7) confirmed no pair of numeric predictors was correlated strongly enough (> 0.7) to raise serious multicollinearity concerns for the Linear Regression model.

7. DATA VISUALISATION

The figures below were selected because each answers a specific question raised during EDA rather than serving as a general chart dump.

7.1 Distribution of Exam Score

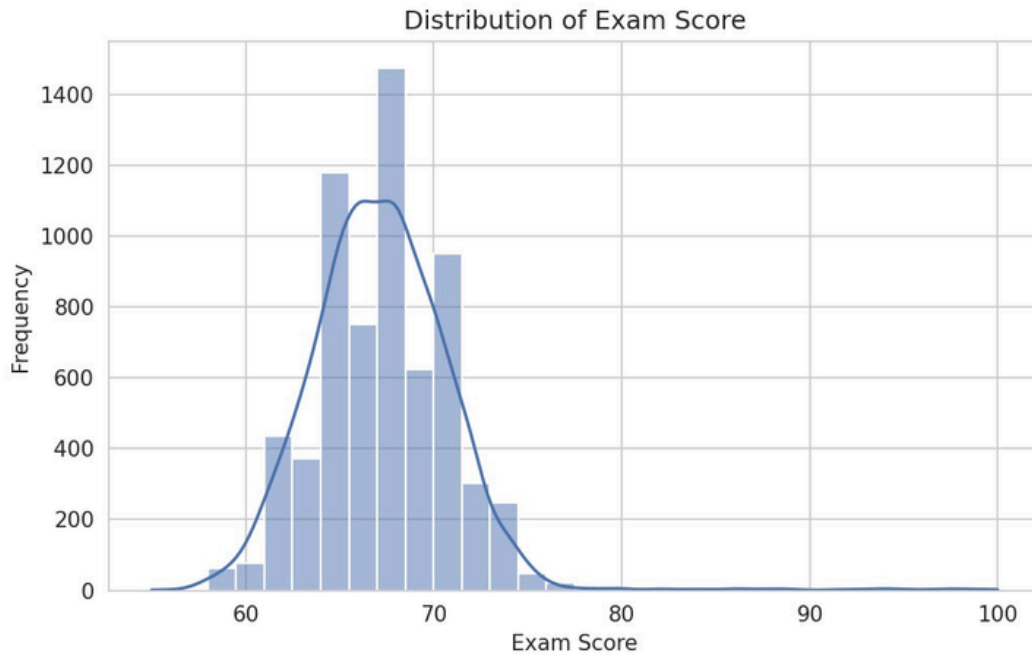


Figure 1. Distribution of Exam Score

The distribution of Exam Score is unimodal and concentrated tightly around the mean of 67.24 (standard deviation 3.89), with most students scoring between roughly 65 and 70. The distribution shows a mild positive skew (skewness = 1.64) caused by a small number of high-performing students in the upper tail; a small number of originally implausible values above 100 were corrected during cleaning (Section 4.4). This confirms Exam Score is well-behaved enough for linear modelling without a transformation.

7.2 Parental Involvement and Exam Score

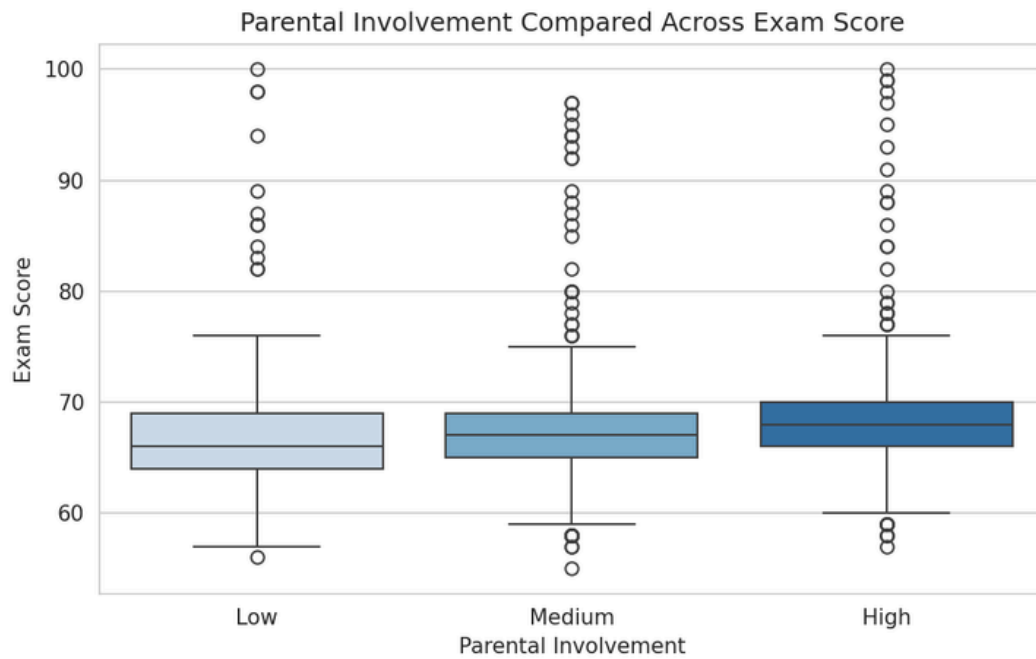


Figure 2. Parental Involvement Compared Across Exam Score

Median Exam Score rises consistently from Low to Medium to High parental involvement, and the interquartile range shifts upward with each level, indicating that students with more actively involved parents tend to achieve modestly higher exam scores on average. This directly answers Research Question 3 for this variable and supports keeping Parental_Involvement as an ordinal feature in the model.

7.3 Peer Influence and Exam Score

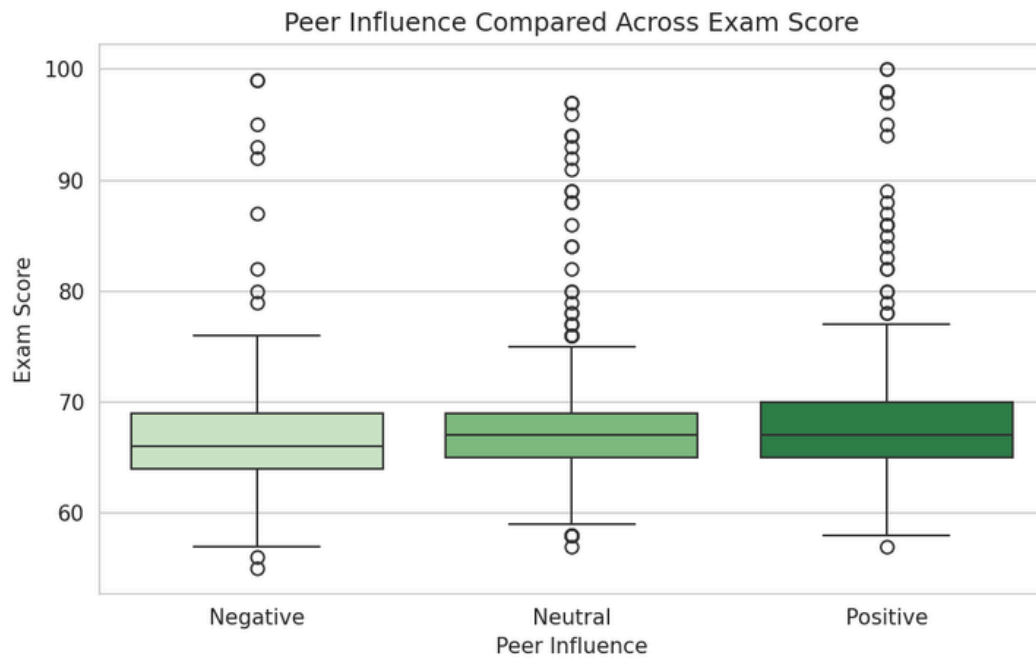


Figure 3. Peer Influence Compared Across Exam Score

A similar, slightly weaker pattern appears for Peer Influence: students reporting a Positive peer influence show a higher median Exam Score than those reporting Neutral or Negative influence, consistent with the idea that social environment contributes to academic outcomes alongside individual study behaviour.

7.4 Hours Studied vs Exam Score

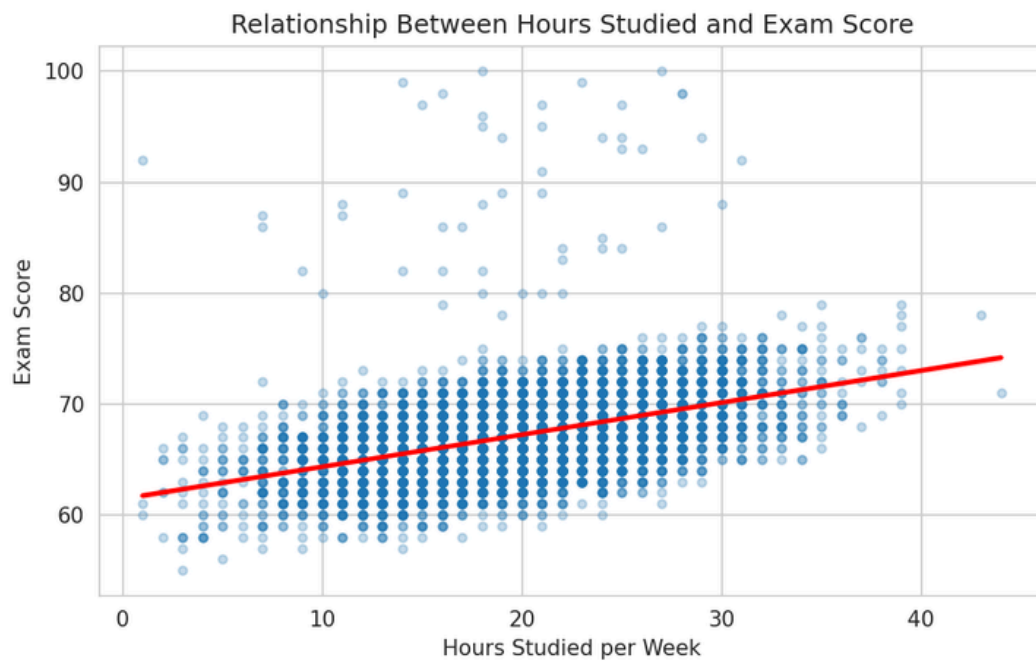


Figure 4. Relationship Between Hours Studied and Exam Score

Hours Studied shows a clear positive linear relationship with Exam Score ($r = 0.446$); students who report studying more hours per week tend to achieve higher exam scores, though with considerable scatter, indicating that study time alone does not fully determine performance. This motivates including Hours_Studied as a direct numeric predictor.

7.5 Attendance vs Exam Score

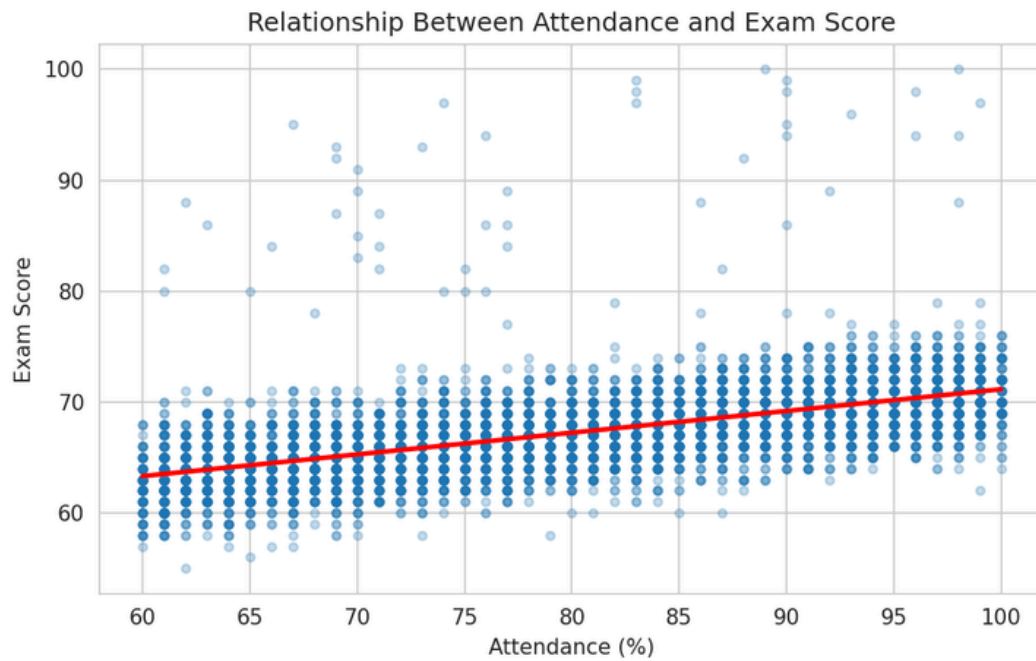


Figure 5. Relationship Between Attendance and Exam Score

Attendance shows the strongest linear association with Exam Score among all numeric variables ($r = 0.581$), stronger than Hours Studied, suggesting that consistent classroom presence may be an even more reliable predictor of performance than self-reported study time. This does not, on its own, establish that attendance causes higher scores — motivated, higher-performing students may simply also attend more consistently (see Section 11, Discussion).

7.6 Correlation Heatmap

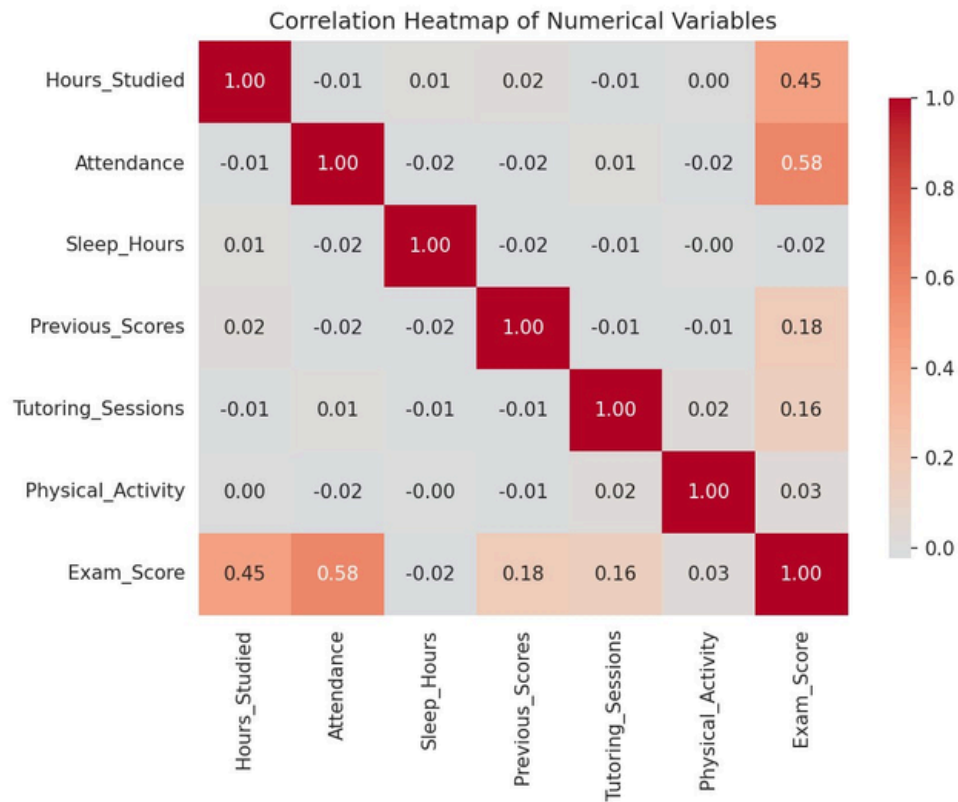


Figure 6. Correlation Heatmap of Numerical Variables

The heatmap consolidates the pairwise relationships explored above: Attendance and Hours Studied stand out as the numeric variables most correlated with Exam Score, while inter-predictor correlations remain modest, supporting the decision to retain all of them as separate features in the model rather than combining or dropping any for redundancy.

8. STATISTICAL ANALYSIS

Descriptive statistics were computed for all numeric variables to summarise central tendency, dispersion, and shape.

Variable	Count	Mean	Std	Min	25%	50%	75%				Max	Skew
Hours_Studied	6607	19.98	5.99	1	16	20	24				44	0.01
Attendance	6607	79.98	11.55	60	70	80	90				100	0.01
Sleep_Hours	6607	7.03	1.47	4	6	7	8				10	-0.02
Previous_Scores	6607	75.07	14.4	50	63	75	88				100	0
Tutoring_Sessions	6607	1.49	1.23	0	1	1	2				8	0.82
Physical_Activity	6607	2.97	1.03	0	2	3	4				6	-0.03
Exam_Score	6607	67.24	3.89	55	65	67	69				100	1.64

Table 8. Descriptive Statistics of Key Numeric Variables

Most behavioural variables (Hours Studied, Attendance, Sleep Hours, Previous Scores) are approximately symmetric (skewness close to 0), so their mean and median are very close and either is a representative measure of centre. Tutoring Sessions is more right-skewed (skew = 0.82), reflecting that most students attend few tutoring sessions while a minority attend many; for this variable the median (1 session) is a more representative 'typical' value than the mean, which is pulled upward by the small number of high-tutoring students. Exam Score is mildly right-skewed after cleaning (skew = 1.64), again meaning the median is a slightly more robust summary of typical performance than the mean, although the two remain close in practice.

It is important to note that the correlations reported in Section 6.2 (e.g. Attendance and Hours Studied with Exam Score) describe association, not causation. Students who attend more and study more may also differ in other unmeasured ways (such as motivation or family support) that independently affect performance; the regression model in Sections 9-10 quantifies predictive association, not a causal effect of any single variable.

9. MACHINE LEARNING METHODOLOGY

9.1 Problem Formulation

This is a supervised regression problem: the goal is to predict a continuous target variable, Exam Score, from a set of numeric and encoded categorical input features (Section 5.5). The final feature set was used as input (X), and Exam_Score as the target (y). No feature directly derived from or revealing the exam outcome was included, so no target leakage is present.

9.2 Feature and Target Preparation

X (features) consists of the 25 variables listed in Section 5.5: original numeric behavioural measures, ordinal-encoded ordered categories, one-hot encoded nominal categories, and the three engineered features. y (target) is Exam_Score. The 80/20 train-test split (Section 5.4) was performed before fitting the StandardScaler, so no information from the test set influenced the scaling parameters.

9.3 Model Selection

Two models were trained and compared. Linear Regression was chosen as an interpretable baseline that directly quantifies each feature's contribution via its coefficient, which is valuable for explaining results to non-technical stakeholders. A Random Forest Regressor was chosen as a non-linear alternative capable of capturing interactions and non-linear effects between features, to test whether flexibility beyond a linear form improves predictive accuracy.

9.4 Training the Model

Key training code for both models is shown below; the full, run-in-order code is provided in the accompanying notebook and in the Appendix.

```
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LinearRegression
from sklearn.ensemble import RandomForestRegressor

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42)
```

```

scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

lr_model = LinearRegression()
lr_model.fit(X_train_scaled, y_train)

rf_model = RandomForestRegressor(
    n_estimators=300, max_depth=8, random_state=42)
rf_model.fit(X_train, y_train)

```

10. MODEL RESULTS AND EVALUATION

10.1 Regression Evaluation

Both models were evaluated on the held-out 20% test set using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R-squared (R²). RMSE and MAE are reported in exam-score points, so both are directly interpretable against the 0-100 score scale.

Model	Dataset	MAE	MSE	RMSE	R-squared
Linear Regression	Train	0.496	4.341	2.084	0.7175
	Test	0.45	3.25	1.803	0.7701
Linear Regression	Test	0.45	3.25	1.585	0.8365
Random Forest	Train	0.95	2.513	2.224	0.65
Random Forest	Test	1.172	4.948		

Table 9. Regression Model Performance (Train vs Test)

On the test set, Linear Regression achieved an R-squared of 0.7701, meaning it explains approximately 77% of the variance in Exam Score, with an average prediction error (MAE) of only 0.45 points — a small, practically acceptable error on a roughly 55-100 point scale. The Random Forest model achieved a lower test R-squared of 0.65 and a higher RMSE of 2.224, indicating that the added flexibility of a non-linear, tree-based model did not translate into better generalisation on this dataset.

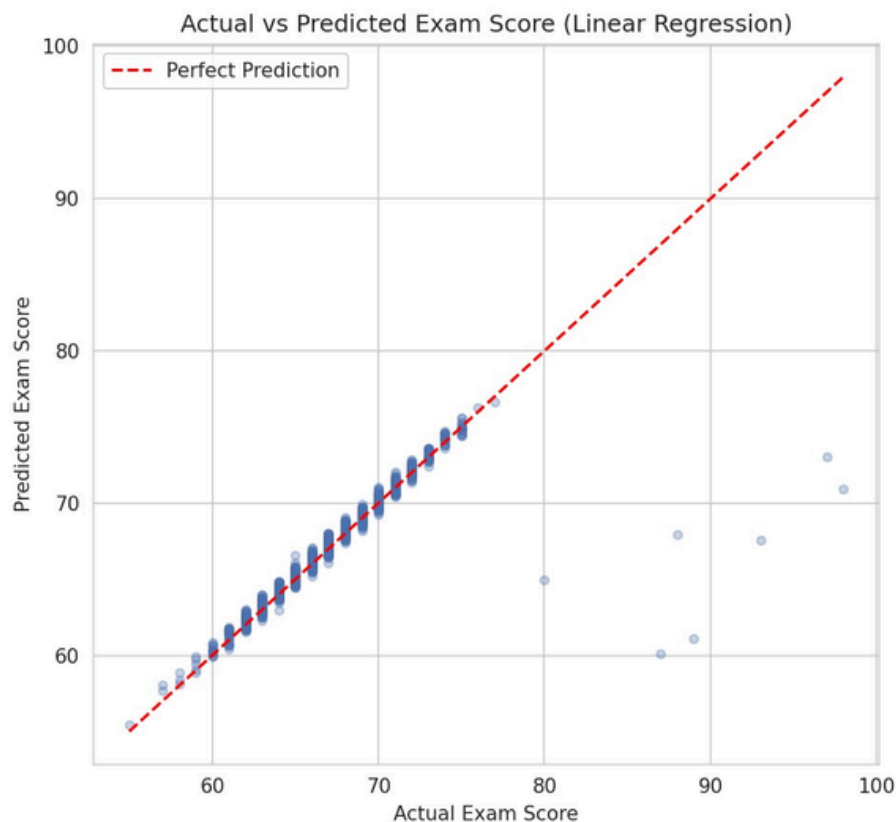


Figure 7. Actual vs Predicted Exam Score (Linear Regression, Test Set)

The scatter of predicted versus actual values clusters tightly around the diagonal line of perfect prediction across the bulk of the score range, confirming the strong test-set R-squared. A small number of points in the upper tail (very high actual scores) are under-predicted, consistent with the mild positive skew identified in Section 8.

10.2 Feature Influence

For Linear Regression, the standardised coefficients below indicate the direction and relative strength of each feature's association with ExamScore, holding other features constant.

Feature	Standardised Coefficient
Attendance	2.138
Engagement_Index	1.118
Peer_Influence_Positive	1.052
Internet_Access_Yes	0.959
Distance_from_Home_Near	0.855
Learning_Disabilities_Yes	-0.855
Access_to_Resources_Ord	0.732
Hours_Studied	0.724
Previous_Scores	0.707
Parental_Involvement_Ord	0.7

Table 10. Top Linear Regression Coefficients

For the Random Forest model, feature importance (based on mean decrease in impurity) is shown below.

Feature	Importance
Attendance	0.3924
Engagement_Index	0.3275
Previous_Scores	0.0647
Study_Sleep_Ratio	0.0299
Parental_Involvement_Ord	0.028
Access_to_Resources_Ord	0.0267
Hours_Studied	0.0248
Physical_Activity	0.0152
Sleep_Hours	0.0104
Family_Income_Ord	0.0102

Table 11. Top Random Forest Feature Importances

Both models agree that Attendance and the engineered Engagement_Index are the dominant predictors of Exam Score, consistent with the correlation analysis in Section 6.2. This convergence across two structurally different models strengthens confidence that these two factors are genuinely the most important drivers of exam performance in this dataset, rather than an artefact of one particular modelling technique.

10.3 Overfitting and Underfitting

Comparing train and test performance: Linear Regression's R-squared changed only marginally between train (0.7175) and test (0.7701), indicating a well-fitted model with no meaningful overfitting. The Random Forest model showed a noticeably larger gap between train R-squared (0.8365) and test R-squared (0.65), a signature of mild overfitting despite the max_depth constraint used to regularise the trees. This further supports selecting Linear Regression as the preferred model for this problem.

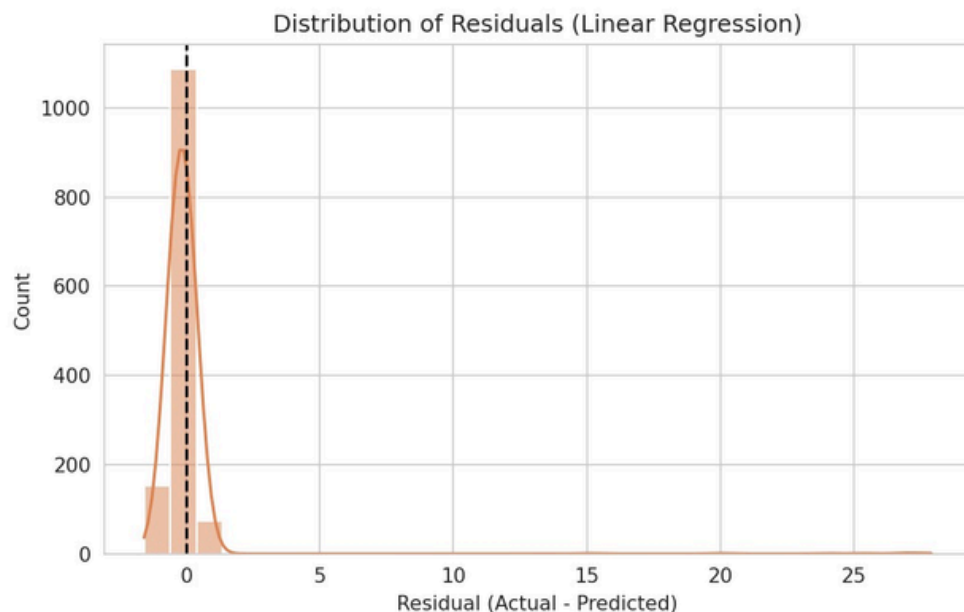


Figure 8. Distribution of Residuals (Linear Regression, Test Set)

The residuals are approximately centred at zero and roughly symmetric, with no strong systematic pattern, supporting the assumption that the linear model's errors are reasonably well-behaved rather than systematically biased.

11. DISCUSSION

The results indicate that a small set of behavioural factors — chiefly attendance and study engagement — account for the majority of explainable variation in student exam performance within this dataset. This is broadly consistent with the educational research literature, which frequently identifies attendance and time-on-task as leading behavioural correlates of achievement. The comparatively weak influence of Sleep Hours and Physical Activity does not necessarily mean these factors are unimportant for wellbeing; it may instead reflect a limited range of variation in the sampled data, or that their effects on performance operate indirectly (for example, through concentration or attendance) rather than directly. The finding that Linear Regression outperformed Random Forest suggests that, for this particular dataset, the dominant relationships are close to linear and additive; a more complex model was not needed to capture the main structure in the data, and in fact generalised slightly worse. As emphasised in Section 8, it is important to stress that all relationships reported here are observed associations from cross-sectional data, not causal effects: for example, students who attend more classes may also be more intrinsically motivated, and it is this underlying motivation — not attendance alone — that could be driving higher scores.

12. LIMITATIONS

- The dataset is cross-sectional (a single snapshot per student), so no causal claims can be made about which factors cause changes in exam performance over time.
- Self-reported behavioural variables (e.g. Hours Studied, Sleep Hours) may be subject to recall or reporting bias, since they were not independently measured.
- The dataset does not include information on subject difficulty, teacher identity, or classroom-level effects, which could also influence exam outcomes but are not captured here.
- Model performance was evaluated on a single train-test split; results could be strengthened with k-fold cross-validation to assess variance in performance estimates.
- The sample, while reasonably large (6,607 records), reflects only the population and context from which it was collected and may not generalise to other schools, regions, or education systems without re-validation.

13. CONCLUSION AND RECOMMENDATIONS

13.1 Conclusion

This project set out to understand and predict student Exam Score using the Student Performance Factors dataset, and to identify which behavioural, familial, and institutional factors matter most. After thorough cleaning (missing value imputation, outlier correction) and preprocessing (encoding, scaling, feature engineering), a Linear Regression model was able to explain approximately 77% of the variance in Exam Score on unseen test data, with an average prediction error of only 0.45 points — outperforming a more complex Random Forest alternative. Attendance and overall engagement (captured through the engineered Engagement_Index) emerged consistently as the strongest predictors across both modelling

approaches, followed by Hours Studied and Previous Scores. These results demonstrate that a compact, interpretable model built on readily available school data can provide meaningful, actionable insight into the drivers of student performance, directly answering the primary prediction question posed in Section 2.2.

13.2 Recommendations

- Attendance is the strongest single predictor of Exam Score (Section 6.2, Table 7); therefore, schools should invest in systems that flag declining attendance early and trigger timely outreach to students and families.
- The Engagement_Index (study hours + attendance + tutoring) ranks among the top predictors in both models (Section 10.2); therefore, schools could bundle these three levers into a single 'engagement' support program rather than addressing them separately.
- Tutoring_Sessions showed a positive correlation with Exam Score ($r = 0.16$, Table 7); therefore, low-cost, targeted tutoring should be prioritised for students with low attendance or low prior scores.
- Parental Involvement and Peer Influence both show a visible positive association with performance (Section 7.2-7.3); therefore, schools could run parent-engagement workshops and peer-mentoring programs to reinforce a supportive academic environment.
- Given its strong test performance and interpretability (Section 10.1), the trained Linear Regression model could be piloted as a lightweight early-warning tool to flag students predicted to score below a target threshold, enabling proactive academic support.

14. REFERENCES

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15. APPENDIX

A. Code Snippets

Full, run-in-order code is provided in the accompanying Jupyter/Colab notebook. The snippets below highlight the key analytical and modelling steps.

A.1 Data Loading and Inspection

```
import pandas as pd
```

```
df = pd.read_csv('P_2_StudentPerformanceFactors.csv')
print(df.shape)
print(df.info())
print(df.isnull().sum())
```

A.2 Data Cleaning

```
df = df.drop_duplicates()
for col in ['Teacher_Quality', 'Parental_Education_Level', 'Distance_from_Home']:
    df[col] = df[col].fillna(df[col].mode()[0])
df['Exam_Score'] = df['Exam_Score'].clip(upper=100)
```

A.3 Preprocessing and Feature Engineering

```
ordinal_map = {'Low':1, 'Medium':2, 'High':3}
for c in ['Parental_Involvement', 'Access_to_Resources', 'Motivation_Level',
          'Family_Income', 'Teacher_Quality']:
    df[c+'_Ord'] = df[c].map(ordinal_map)

df['Engagement_Index'] = df['Hours_Studied'] + df['Attendance']/10 +
df['Tutoring_Sessions']*2

nominal_cols = ['School_Type', 'Peer_Influence', 'Gender', 'Internet_Access',
                'Learning_Disabilities', 'Distance_from_Home',
                'Parental_Education_Level', 'Extracurricular_Activities']
df = pd.get_dummies(df, columns=nominal_cols, drop_first=True)
```

A.4 Model Training and Evaluation

```
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
import numpy as np

pred = lr_model.predict(X_test_scaled)
mae = mean_absolute_error(y_test, pred)
rmse = np.sqrt(mean_squared_error(y_test, pred))
r2 = r2_score(y_test, pred)
print(f'MAE={mae:.3f} RMSE={rmse:.3f} R2={r2:.4f}')
```

B. Additional Figures and Tables

Supplementary diagnostic figure reproduced here for reference (duplicated from Figure 8).

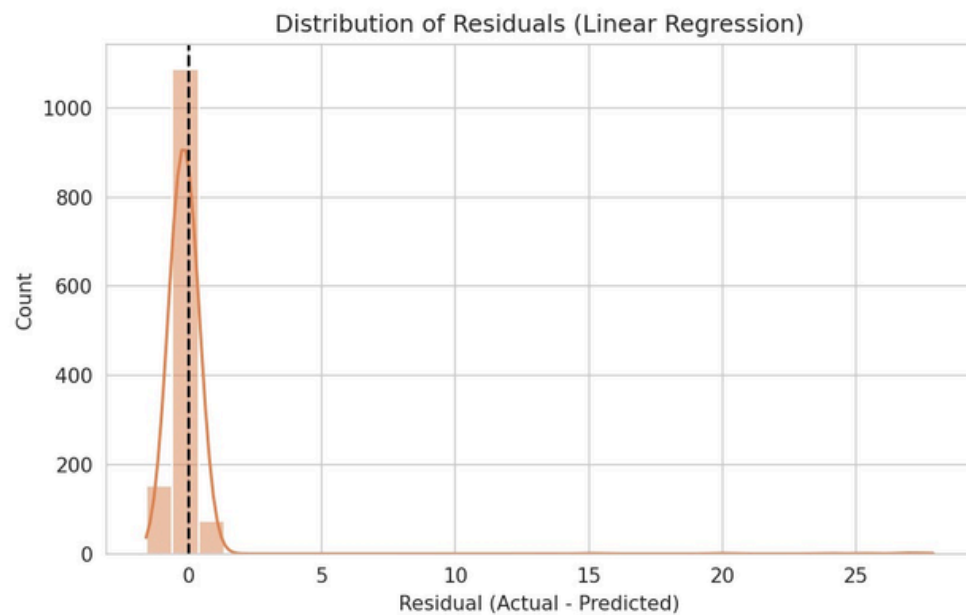


Figure B1. Residual Distribution (Linear Regression, Test Set)